Métodos Matemáticos de la Física II

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Bla bla bla... resumen...

I. METRIC SPACE

A set X is called a *metric space* if it is equiped with a notion of *metric distance*, i.e. a function $d : X \times X \to \mathbb{R}$ such that, for any three elements x, y and z in X, which are called *points*, it holds:

1. $d(x, y) \geq 0$, i.e. it is *non-negative*.

2. $d(x, y) = 0$ implies $x = y$, i.e. it is *positive definite*.

3. $d(x, y) = d(y, x)$, i.e. it is *symmetric*.

4. $d(x, z) \leq d(x, y) + d(y, z)$, i.e. it satisfies the *triangular inequality*.

Example: The set $\mathbb{R}^n$ with the euclidean distance $d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$ is a metric space.

A. Open ball

The set

$$B_r(x) = \{y \in \mathbb{R}^n : d(x, y) < r\}$$

is called the *open ball* of $\mathbb{R}^n$ centered at x and of radious $r > 0$.

B. Open set

A subset U of $\mathbb{R}^n$ is *open* if for every $x \in U$ there exists an open ball $B_r(x) \subset U$.

C. Complement set

Consider two subsets X and Y of $\mathbb{R}^n$ such that $X \subset Y$. The *complement* $X \backslash Y$ of X in Y is the subset of $\mathbb{R}^n$ given by

$$X \backslash Y := \{y \in Y : y \notin X\}.$$

In particular, we can think of the complement $X \backslash \mathbb{R}^n$.

D. Closed set

A subset F of $\mathbb{R}^n$ is *closed* if its complement $F \backslash \mathbb{R}^n$ is open.

E. Limit or accumulation point

A point x is a *limit point* (or *accumulation point*) of a subset A of $\mathbb{R}^n$ if, for every $r > 0$, it holds that

$$(B_r(x)\backslash\{x\}) \cap A \neq \emptyset.$$

In other words, there are points in A that are arbitrarily close to x.

Example: The points 0 and 1 are limit points of the open interval $(0, 1)$.

F. Closure set

The *closure* $\overline{A}$ of a subset A of $\mathbb{R}^n$ is defined by

$$\overline{A} = A \cup \{\text{all limiting points of } A\}.$$

Examples: the closed interval $[0, 1]$ is the closure of itself and the subsets $(0, 1)$, $[0, 1)$, $(0, 1]$ and $(0, 1/2) \cup (1/2, 1)$.

G. Bounded set

A subset A of $\mathbb{R}^n$ is *bounded* if there exists a finite $r > 0$ such that $B_r(0) \cap A = A$.

H. Compact set

A subset K of $\mathbb{R}^n$ is *compact* if K is closed and bounded.

II. TOPOLOGICAL SPACE

Previous definitions of open and closed sets depend on the existence of a metric distance. It is possible to generalize the formalism using a weaker, i.e. more general, requirement. The notions of open and closed sets can be defined when the set is equipped with a topology, which provides a weaker notion of space, than that of a metric space. Informally, a topology is essentially the structure that specifies:

- which sets are open,
- which sequences converge in a given set, and
- which functions are continuous over such set.

A *topological space* is a set X equipped with a collection T of subsets of X called *open* sets that satisfy the following properties:

1. X and $\emptyset$ belong to T.
2. Arbitrary unions of open sets are open. Formally, for any $R \subset T$, then
$$\left(\bigcup_{U\in R} U\right) \in T.$$
3. Finite intersections of open sets are open. Formally, for any $R \subset T$ such that $|R| \in \mathbb{N}$, then
$$\left(\bigcap_{U\in R} U\right) \in T.$$

In other words, the notion of openness is a primite definition.

The notions of *complement* set and *closed* set are analogous to those previously defined. Namely, the complement of a subset S of X is the subset $S\backslash X := \{x \in X : x \notin S\}$ of X, and a subset Y of X is closed if its complement $Y\backslash X$ is open.

A. Closure set

The *closure* $\overline{A}$ of a subset A of X is defined as the intersection of all closed subsets of X that contain A. Formally,

$$\overline{A} = \bigcap_{F: F\subset X,\, A\subset F,\, F \text{ is closed}} F.$$

B. Limit point

A point x is a *limit point* (or *accumulation point*) of a subset A of X if, for each open subset U of X such that $x \in U$, then

$$(U\backslash\{x\}) \cap A \neq \emptyset.$$

This is the purely topological generalization of the notion defined in section I E.

C. Open cover

A family of open sets $\{U_i : i \in I\} \subset \mathcal{T}$ is called an *open cover* of K if

$$K \subset \bigcup_{i\in I} U_i.$$

D. Compact set

A subset K of X is *compact* if for every open cover $\{U_i : i \in I\}$ of K, there exists a finite subset

$$\{i_1, \dots, i_n\} \subset I$$

such that

$$K \subset U_{i_1} \cup \cdots \cup U_{i_n}.$$

The finite family

$$U_{i_1}, \dots, U_{i_n}$$

is called a *finite subcover*.

Example: The family

$$U_n = \left(\frac{1}{n}, 1\right), \qquad n \in \mathbb{N},$$

is an open cover of $(0, 1)$ because

$$(0,1) = \bigcup_{n=1}^{\infty} \left(\frac{1}{n}, 1\right),$$

but no finite subcollection of it covers $(0, 1)$. Namely, if we choose finitely many:

$$U_{n_1}, \dots, U_{n_k}.$$

then

$$U_{n_1} \cup \cdots \cup U_{n_k} = \left(\frac{1}{N}, 1\right).$$

with $N = \max(n_1, \dots, n_k)$, missing all points in $(0, 1/N]$. Hence, the finite union does not cover $(0, 1)$. Thus $(0, 1)$ admits an open cover with no finite subcover, and therefore it is not compact.

III. TEST FUNCTIONS

A. Support of a function

The *support* of a function $\varphi : \mathbb{R}^n \to \mathbb{R}$, is the subset K of $\mathbb{R}^n$ given by:

$$\mathrm{supp}(\varphi) = \overline{\{x \in \mathbb{R}^n : \varphi(x) \neq 0\}}.$$

In other words, it is the closure of the complement of the kernel of φ.

B. Smooth function

Let $C^\infty(\mathbb{R}^n)$ denote the set of *smooth functions* (i.e. of infinitelly differentiable functions) from $\mathbb{R}^n$ to $\mathbb{R}$.

C. Space of compactly supported smooth test functions

The (vector) *space of smooth compactly supported functions* $C_c^\infty(\mathbb{R}^n) = \mathcal{D}(\mathbb{R}^n)$ is such that, if $\varphi \in C_c^\infty(\mathbb{R}^n)$, then:

1. $\varphi : \mathbb{R}^n \to \mathbb{R}$.
2. $\varphi \in C^\infty(\mathbb{R}^n)$, i.e. it is infinitely differentiable.
3. The support $\mathrm{supp}(\varphi)$ is compact.

The members of $C_c^\infty(\mathbb{R}^n)$ are called *compactly supported smooth test functions.*

Excercise: Show that the linear combination and the product of compactly supported smooth test functions are also compactly supported smooth test functions.

D. Schwartz space of functions

The *Schwartz (vector) space of functions* $\mathcal{S}(\mathbb{R}^n)$ is such that, if $\varphi \in \mathcal{S}(\mathbb{R}^n)$, then:

1. $\varphi : \mathbb{R}^n \to \mathbb{R}$.
2. $\varphi \in C^\infty(\mathbb{R}^n)$, i.e. it is infinitely differentiable.
3. For every $\alpha_1, ..., \alpha_n, \beta_1, ..., \beta_n \in \{0, 1, 2, ...\}$, it holds that

   $$\mathrm{supr}_{x \in \mathbb{R}^n} \left| x_1^{\alpha_1} ... x_n^{\alpha_n} \partial_1^{\beta_1} ... \partial_n^{\beta_n} \varphi(x) \right| < \infty.$$

   In other words, all the derivatives of φ multiplied by arbitrary powers of the components of x are bounded in $\mathbb{R}^n$.

The members of $\mathcal{S}(\mathbb{R}^n)$ are called *Schwartz test functions.*

Excercise: Show that the linear combination and the product of Schwartz test functions are also Schwartz test functions.

IV. TOPOLOGIES ON THE SPACES OF TEST FUNCTIONS

To eventually define distributions, a notion of convergence should be introduced to each of the spaces of test functions. More specifically, the uniform converge of test functions is required, because it gives strong control over functions and their derivatives.

A. Review on the convergence of sequences

To eventually define the notion of a continuous distribution, it is useful to review the notion of continuity of standard functions. The continuity of a function $f : \mathbb{R}^n \to \mathbb{R}$ can be defined in terms of sequences. Therefore, it makes sense to review first the notion of the convergence of a sequence.

When a sequence $\{x_j : j \in \mathbb{N}\}$ converges to x we say that $x_j \to x$ when $j \to \infty$. Formally, this means that, for any $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that $|x_j - x| < \epsilon$ for every $j \geq N$.

B. The topology of $C_c^\infty(\mathbb{R}^n)$

We say that a sequence of test functions $\{\varphi_j \in C_c^\infty(\mathbb{R}^n) : j \in \mathbb{N}\}$ converges to a test function $\varphi \in C_c^\infty(\mathbb{R}^n)$, i.e. $\varphi_j \to \varphi$ when $j \to \infty$, if:

1. the supports of φ and each φ_j lies within a common compact subset K of $\mathbb{R}^n$, and
2. for any $\alpha_1, ..., \alpha_n \in \{0, 1, 2, ...\}$, it holds that

$$\lim_{j\to\infty} \operatorname*{supr}_{x\in K} |\partial^{\alpha_1} ... \partial^{\alpha_n} \varphi_j(x) - \partial^{\alpha_1} ... \partial^{\alpha_n} \varphi(x)| = 0.$$

   In other words, that every derivative converges uniformly on K.

C. The topology of $\mathcal{D}(\mathbb{R}^n)$

We say that a sequence of test functions $\{\varphi_j \in \mathcal{S}(\mathbb{R}^n) : j \in \mathbb{N}\}$ converges to a test function $\varphi \in \mathcal{S}(\mathbb{R}^n)$, i.e. $\varphi_j \to \varphi$ when $j \to \infty$, if for any $\alpha_1, ..., \alpha_n, \beta_1, ..., \beta_n \in \{0, 1, 2, ...\}$, then

$$\lim_{j\to\infty} \operatorname*{supr}_{x\in\mathbb{R}^n} \left| x_1^{\beta_1} ... x_n^{\beta_n} \partial^{\alpha_1} ... \partial^{\alpha_n} (\varphi_j - \varphi)(x) \right| = 0.$$

V. DISTRIBUTIONS

A. Functionals

Consider a space of test functions $\mathcal{F}(\mathbb{R}^n)$. Here, $\mathcal{F}$ may denote, for example, C_c^∞ or $\mathcal{S}$. A *functional* on $\mathcal{F}$ is a function $\Phi : \mathcal{F}(\mathbb{R}^n) \to \mathbb{R}$. Hence, for any test function $\varphi \in \mathcal{F}(\mathbb{R}^n)$, the evaluation $\Phi(\varphi)$ is just a number in $\mathbb{R}$.

B. Review on function continuity

We say that a function $f : \mathbb{R}^n \to \mathbb{R}$ is continuous at $x \in \mathbb{R}^n$ if for any sequence $\{x_j \in \mathbb{R}^n : j \in \mathbb{N}\}$ that converges to x and for any $\epsilon > 0$, there is an $N \in \mathbb{N}$ such that $|f(x_j) - f(x)| < \epsilon$ for every $j \geq N$.

C. Continuous functional

A functional $\Phi : \mathcal{F}(\mathbb{R}^n) \to \mathbb{R}$ is continuous at $\varphi \in \mathcal{F}(\mathbb{R}^n)$ if, for any sequence of test functions $\{\varphi_j \in \mathcal{F}(\mathbb{R}^n) : j \in \mathbb{N}\}$ that converges to some test function $\varphi \in \mathcal{F}(\mathbb{R}^n)$, i.e. $\varphi_j \to \varphi$ as $j \to \infty$, then

$$\Phi(\varphi_j) \to \Phi(\varphi)$$

as $j \to \infty$. Notice that $\Phi(\varphi_j) \to \Phi(\varphi)$ denotes a convergence of numbers. In other words, we are saying that Φ is continuous in φ if

$$\varphi_j \to \varphi \quad \Rightarrow \quad \Phi(\varphi_j) \to \Phi(\varphi).$$

Of course, the convergence of $\varphi_j \to \varphi$ is defined in terms of the particular choice of $\mathcal{F}(\mathbb{R}^n)$.

D. Distribution

A functional $\Phi : \mathcal{F}(\mathbb{R}^n) \to \mathbb{R}$ is a *distribution* if:

1. it is continuous at each $\varphi \in \mathcal{F}(\mathbb{R}^n)$, and
2. it is linear.

Excercise: Show that if $\Phi : \mathcal{F}(\mathbb{R}^n) \to \mathbb{R}$ is a linear functional that is continuous at $\varphi = 0 \in \mathcal{F}(\mathbb{R}^n)$, then it is continuous at every $\varphi \in \mathcal{F}(\mathbb{R}^n)$.